PREMIER UNIVERSITY,CHITTAGONG

Department of Computer Science & Engineering

PROJECT PROPOSAL

Project Name : Smart Campus Network Design

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Introduction

Traditional university setups often face challenges such as inefficient communication, limited centralized monitoring, and lack of robust network management. A well-designed network infrastructure can significantly improve connectivity, security, and operational efficiency across campus.

This project aims to design and simulate a Smart University Campus Network using Cisco Packet Tracer. The focus is primarily on building a fully functional network infrastructure, which includes:

1. **Network Components:** Routers, switches, access points, servers, and end devices (PCs, laptops, printers) to provide seamless internet access, inter-department communication, secure file sharing, and server-based operations such as DNS, DHCP, and web services.
2. **Minimal IoT Integration:** RFID-based access control is implemented to provide secure, touchless entry and attendance tracking, demonstrating how small IoT components can complement the network infrastructure.

Objectives

The main objectives of this project are:

- To design a complete smart university campus network using Cisco Packet Tracer, focusing primarily on traditional networking components.
- To implement a structured and scalable network with routers, switches, servers, and end devices to ensure reliable interconnectivity and internet access.
- To integrate RFID-based access control for secure and touchless entry, as well as automated attendance tracking.
- To simulate real-world scenarios demonstrating efficient network operation and secure user access**.**
- To enhance understanding of network configuration, IP addressing, VLANs, routing protocols, and basic IoT integration.
- To demonstrate how a robust network can support automation and security features in a modern campus environment.

Scope

This project involves designing and simulating a smart university campus network using Cisco Packet Tracer. The scope includes:

- **Campus Coverage:**
  - A single academic building comprising classrooms, laboratories, and administrative offices.
  - Wired and wireless network connectivity across all areas to ensure reliable access for all users.
- **Network Services:**
  - Implementation of essential services such as DHCP, DNS, and web hosting within the network.
  - Structured IP addressing and VLAN configuration for efficient traffic management and enhanced security.
- **IoT Feature:**
  - Integration of RFID-based access control for secure entry and automated attendance tracking, demonstrating how a network can support IoT functionality.

Tools and Technologies

- **Cisco Packet Tracer**

  The primary simulation software used to design, configure, and test the smart campus network, including network services and limited IoT integration (RFID).

- **Networking Devices:**
  - **Routers:** For inter-VLAN routing and network segmentation.
  - **Switches (Layer 2 & Layer 3):** To connect devices and manage VLANs.
  - **Wireless Access Points:** To provide campus-wide Wi-Fi coverage.
  - **PCs and Laptops:** Simulated end-user devices for students, faculty, and administration.
  - **Servers:** Including DHCP, DNS, and Web servers to provide essential network services.

- **IoT Devices:**
  - **RFID Readers & Cards:** Used for secure access control and automated attendance logging.
  - **Smart Doors:** Automatically open/close with RFID verification.
- **Protocols and Services:**
  - **DHCP:** For dynamic IP address assignment.
  - **DNS:** For domain name resolution within the campus network.
  - **HTTP:** To host internal web pages, monitoring dashboards, and RFID access logs.
  - **ACLs (Access Control Lists):** To manage network security and restrict unauthorized access.

Methodology and Implementation

Methodology

The project follows a structured approach to design, configure, and simulate a smart university campus network using Cisco Packet Tracer. The methodology includes:

1. **Requirement Analysis:**
   - Identify the networking needs of the campus, including routers, switches, access points, servers, and end devices.
   - Determine essential services, security requirements, and areas for integration of RFID access.
2. **Network Design:**
   - Plan the physical and logical topology of the campus network.
   - Decide on IP addressing schemes, VLAN segmentation, routing protocols, and wireless coverage.
3. **Device Configuration:**
   - Configure routers and switches with inter-VLAN routing, VLAN assignments, and security features such as ACLs.
   - Set up DHCP and DNS servers for automatic IP assignment and domain name resolution.
   - Deploy a web server for internal monitoring and management dashboards.

4. **RFID Integration:**

   - Integrate RFID readers with smart doors for secure, automated access and attendance tracking.
   - Ensure proper network connectivity between RFID devices and the monitoring server.

5. **Security and Automation:**

   - Implement ACLs, VLANs, and IP-based restrictions to maintain network security.
   - Configure event-driven actions in the network for RFID-based entry logging.

6. **Testing and Validation:**

   - Simulate real-world scenarios such as user access through RFID.
   - Monitor network traffic, device connectivity, and service availability to ensure reliability.

7. **Documentation:**

   - Record all network configurations, routing setups, VLAN assignments, and RFID integration steps for reporting and future reference.

Implementation

1. **Network Infrastructure:**

   - **Routers:** Configured for inter-VLAN routing and traffic segmentation between academic, administrative, and RFID devices.
   - **Switches (Layer 2 & 3):** Connect devices, enforce VLAN separation, and manage traffic efficiently.
   - **Wireless Access Points:** Provide Wi-Fi coverage across the campus, supporting network connectivity for all devices.

2. **IP Addressing and VLANs:**

   - Apply a structured IP addressing scheme for each subnet.
   - Create VLANs to separate traffic, improve security, and optimize performance.

3. **Network Services:**

   - **DHCP:** Assign IP addresses dynamically to end devices.
   - **DNS:** Resolve local domain names for internal services.
   - **Web Server:** Host monitoring dashboards for RFID and network management.

4. **RFID Configuration:**

   - RFID readers authenticate users for smart doors.
   - Attendance and access logs are sent to the server for monitoring.

5. **Security and Access Control:**

   - Use ACLs to restrict unauthorized access to network resources.
   - Monitor traffic for anomalies or failed access attempts.

6. **Simulation and Testing:**

   - Test RFID entry for authorized and unauthorized users.
   - Monitor network stability, service availability, and device communication.
   - Ensure VLAN separation and ACL enforcement work correctly across all network segments.

Project Timeline

The smart university network will be developed over 7 weeks, as outlined below:

Week	Milestones
Week 1	Analyze requirements and finalize network components and project scope.
Week 2	Design the physical and logical topology of the university network in Cisco Packet Tracer.
Week 3	Set up the basic network infrastructure including routers, Layer 2 & Layer 3 switches, PCs, servers, and wireless access points.
Week 4	Implement essential network services such as DHCP, DNS, and web server; configure VLANs and routing protocols.
Week 5	Integrate RFID readers and smart doors for secure access and attendance logging.
Week 6	Perform testing of individual network components, RFID communication, and access control functionality.
Week 7	Conduct final testing, verify network reliability and security, complete documentation, and prepare for project demonstration.

Expected Outcomes

By the end of this project, the following outcomes are expected:

- A fully functional and simulated university network infrastructure in Cisco Packet Tracer, including routers, switches, servers, and access points.
- Integration of RFID readers and smart doors for secure access control and automated attendance logging.
- Implementation of essential network services such as DHCP, DNS, and web server for reliable communication and resource management.
- Verification of network security, VLAN segmentation, and inter-department connectivity.
- Enhanced understanding of how traditional networking and selected IoT devices (RFID-based access) can work together to create a secure, automated, and efficient campus environment.

Risk and Mitigation

Risks

1. **Network Configuration Errors:** Misconfigurations in routers, switches, or VLANs could disrupt network communication and connectivity.
2. **RFID Device Integration Issues:** Failures in RFID readers or communication with smart doors could affect access control and attendance logging.
3. **Simulator Limitations:** Cisco Packet Tracer may not fully support all network features or advanced RFID functionalities.

Mitigations

1. **Incremental Testing and Backup:** Perform stepwise testing of network configurations and maintain backups to quickly recover from errors.
2. **Thorough Device Testing:** Test RFID device communication and integration with the network to ensure reliable access control.
3. **Scope Adjustment:** Focus on supported network and RFID features within Packet Tracer and apply simplified alternatives where necessary.

Limitations

While the project provides a solid foundation, it has certain limitations:

- **Simulation Constraints:** Cisco Packet Tracer supports only a limited set of network and RFID functionalities, which may not fully represent real-world capabilities.
- **No Real-Time Hardware Integration:** The project is limited to a virtual environment; no physical network or RFID devices are used.
- **Simplified Logic:** Network automation and RFID access control logic are basic and may not reflect complex real-world scenarios.
- **Security Limitations:** Advanced cybersecurity measures, such as intrusion detection or real-time threat monitoring, are not implemented.
- **Scalability:** The network design is based on a single-campus simulation and may not directly scale to larger or multi-campus environments.

Complex Engineering Problem Attributes

P1: Depth of Knowledge

This project involves advanced concepts from computer networking, cybersecurity, and networked device integration. It requires a solid understanding of protocols such as DHCP, DNS, HTTP, and routing configurations, as well as the setup and management of VLANs, subnets, and access control mechanisms like RFID.

P2: Wide-Ranging or Conflicting Issues

Designing a campus network with multiple departments and secure access control introduced challenges where requirements often conflicted ,for example, ensuring high network performance while maintaining secure RFID-based entry.

P3: Abstract Thinking & Analysis

The system required simulating a real world university environment using Cisco Packet Tracer. This meant abstracting the physical infrastructure, user behavior, and network traffic into a virtual model and logically predicting interactions and outcomes.

P4: Infrequently Encountered Issues

Combining a full-scale enterprise level network with access control mechanisms in a simulation is rarely done at the undergraduate level. Unique features like RFID-based automated entry added to the complexity.

P5: Adherence to Standards

The project follows well established networking standards, such as VLAN configuration, IP addressing schemes, ACLs, and secure authentication protocols, ensuring the design aligns with industry practices.

P6: Stakeholder Conflicts

Multiple user groups had differing needs,students expected reliable connectivity, security teams required secure access control, and administrators needed a centralized, manageable system. Balancing these requirements added complexity.

P7: Interdependent Sub-Problems

Each part of the system is tightly connected. For example, the RFID access system depends on proper network configuration and server availability. A fault in one part can affect overall network performance and security.

Conclusion

This project focuses on designing a smart university campus by emphasizing a robust and secure network infrastructure using Cisco Packet Tracer. The project demonstrates the implementation of a structured campus network with routers, switches, VLANs, servers, and wireless access points, ensuring reliable connectivity, inter-department communication, and efficient management of network resources.

RFID-based access control is integrated to provide secure and automated entry for students and staff, highlighting how networked systems can enhance campus security. Through this project, hands-on experience is gained in network configuration, VLAN management, IP addressing, server setup, and access control, offering a practical understanding of building a secure, scalable, and intelligent campus network.